BUS-F420: Equity and Fixed Income Investments

Lecture 14: Measuring Interest Rate Risk

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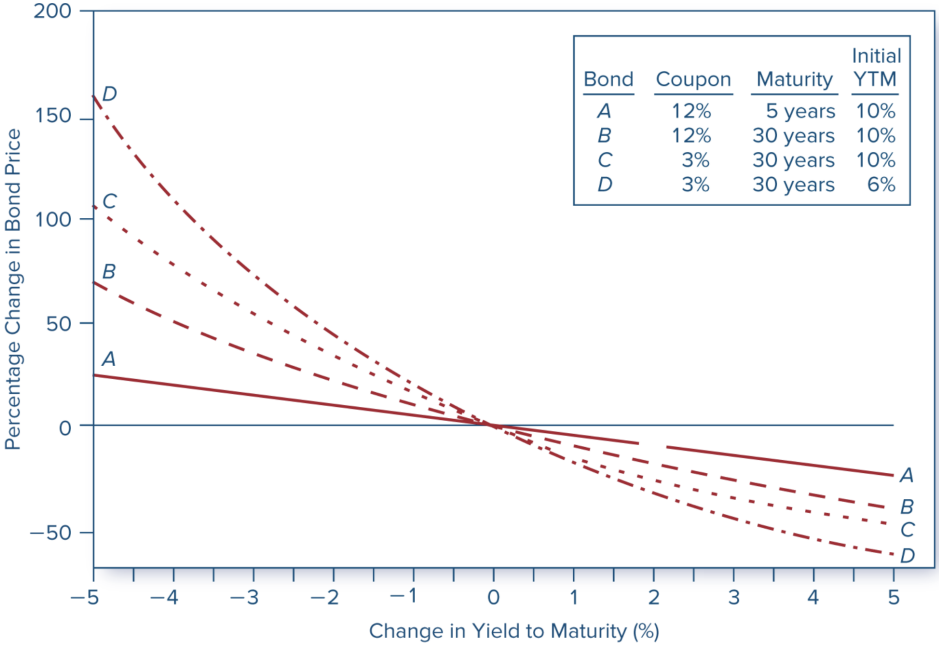
Recap

- Last time, focused on understanding **yields to maturity**
  - Discount rate consistent with a given bond price
  - Encompasses coupons, capital gains, and reinvestment

- **Key problem:** Depended on holding bonds until maturity! Not always the case
  - If market yields change, prices change $\rightarrow$ short-term capital gains
  - Important for bonds traders, regulators, etc.

- **Today:** Characterize *how* prices change with market yields


Bond | Coupon | Maturity | Initial YTM
A | 12% | 5 years | 10%
B | 12% | 30 years | 10%
C | 3% | 30 years | 10%
D | 3% | 30 years | 6%
Percentage Change in Bond Price
200
150
100
50
0
−50
D
C
B
A
−5
−4
−3
−2
−1
0
1
2
3
4
5
Change in Yield to Maturity (%)

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Rule 3: Price Sensitivity is Higher for Long Maturity Bonds

- Let's examine this rule in the context of a zero coupon bond with:
  1. ...a maturity of T years
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- **What's the intuition for this result?**
  - Higher maturity $\rightarrow$ *more compounding*

Rule 5: Low Coupon Bonds Have Greater Interest Rate Risk

- Consider a bond that pays a coupon at time 1 and only pays par value at time 2:

$$P = \frac{\textbf{Coupon}}{1+y} + \frac{M}{(1+y)^2}$$

- Now, calculation becomes less trivial. Will be useful to define the following **weights**:

$$w_1 = \frac{\textbf{Coupon}}{(1+y)P}, \qquad w_2 = \frac{M}{(1+y)^2 P}$$

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  - What do the weights w_1 and w_2 add up to? 1!
  - If the bond were actually a zero, what would w_2 be? 1!
  - What happens to w_2 if the coupon increases? $\downarrow$!

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- It turns out that the sensitivity to interest rates can be written as

$$\frac{dP}{P} = -\underbrace{\left(1 \times w_1 + 2 \times w_2\right)}_{\text{weighted average of payment maturities}} \times \frac{dy}{1+y}$$

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- **What's the intuition for this result?**
  - The higher is the **coupon** payment, the lower the effective maturity!
  - *Conversely:* the <u>lower</u> is the coupon payment, the more weight put on par value

- This weighted *effective* maturity is called **duration**

Duration for a General Option-Free Bond

- For an option-free bond with cash flows CF_t, **(modified) duration** is defined as

$$\boldsymbol{D} = \frac{1}{1+y}\sum_{t=1}^{T} \boldsymbol{w_t t}, \qquad \boldsymbol{w_t} = \frac{\boldsymbol{CF_t}}{(1+y)^t P}$$

- Duration takes each cash flow's maturity into account, which implies (all else equal)
  - Longer maturity bonds have *higher* duration
  - Higher coupon bonds have *lower* duration

Note: If using *semi-annual* cash flows, must divide by 2 to calculate *annual* duration

How is Duration Related to Interest Rate Risk?

- For a general bond with generic cash flows, the price effect of a **small** change in yield is

$$\frac{\Delta P}{P} = -\boldsymbol{D} \times \Delta y$$

- **Duration** allows us to <u>combine</u> all aspects of a bond into a single measure

Implications

- Two bonds with identical time to maturity can have different duration
- Two bonds with identical coupons can have different duration
- Two bonds with identical time to maturity *and* coupons can have different duration

Calculating Modified Duration

3-year corporate bond, 10% annual coupon, 8% annual yield, par value $1,000

Period	Cash Flow	Discounted Cash Flow	Weight	Weighted Maturity
1				
2				
3				
4				
5				
6				

Annual Modified Duration:

Calculating Modified Duration

3-year corporate bond, 10% annual coupon, 8% annual yield, par value $1,000

Period	Cash Flow	Discounted Cash Flow	Weight	Weighted Maturity
1	$50			
2	$50			
3	$50			
4	$50			
5	$50			
6	$1050			

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Period	Cash Flow	Discounted Cash Flow	Weight	Weighted Maturity
1	$50	$48.08		
2	$50	$46.23		
3	$50	$44.45		
4	$50	$42.74		
5	$50	$41.10		
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6	$1050	$829.83	0.788	4.731
		Annual Modified Duration:		**2.57**

Verifying Interest Rate Sensitivity Rules

Bond	Maturity	Coupon Rate	Yield	Par Value	Duration
A	5	12%	10%	$1,000	
B	30	12%	10%	$1,000	
C	30	3%	10%	$1,000	
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D	30	3%	6%	$1,000	**16.06**

Does Duration Change, or is it a fixed characteristic?

- **C** $\rightarrow$ **D: Duration** may *change* with market conditions

- Let's think about this with a basic T-year zero:

$$\boldsymbol{D} = \frac{\boldsymbol{T}}{1+y}$$

- If $\boldsymbol{T} = 20$, what is the duration for different y values?
  - $y = 10\%$

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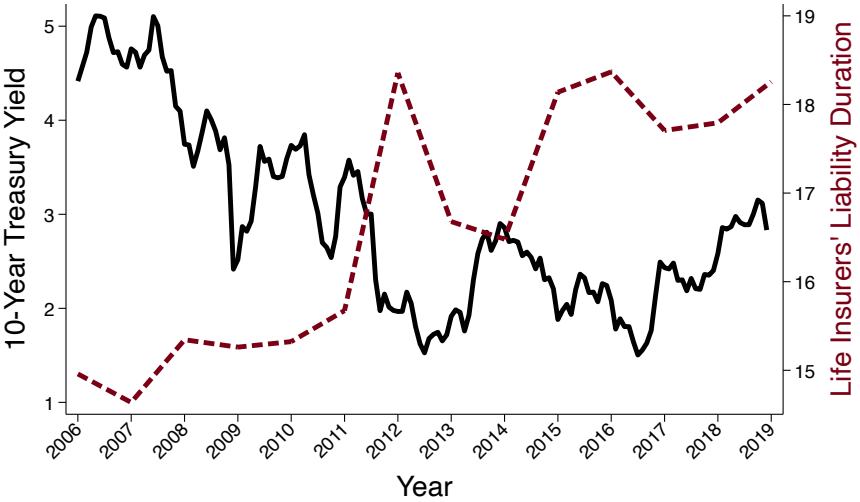
  $\rightarrow$ For a 0.5% decline in yields, $18.2 \times 0.5 = 9.1\%$ vs. $19.4 \times 0.5 = 9.7\%$ increase in price!

Application: Yields and the Duration of Life Insurance Liabilities



A Simpler Way to Calculate Duration

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- **Instead:** Approximate duration directly by computing prices for *a 1bp change in yields:*

$$\underbrace{\text{Present Value of a Basis Point}}_{\equiv \text{PVBP}} = \frac{P^+ - P^-}{2}, \qquad \textbf{Effective Duration} = -\frac{PVBP}{P_0 \times 0.0001}$$

where P^+ is calculated with yield $y + 0.0001$
P^- is calculated with yield $y - 0.0001$

Example Using $\Delta y = 0.0001$

	A (Baseline)	B (Maturity ↑)	C (Coupon ↓)	D (Yields ↓)
True duration	3.76	9.29	11.59	16.06
P^+				
P_0				
P^-				
PVBP				
Eff. Duration				

Example Using $\Delta y = 0.0001$

	A (Baseline)	B (Maturity ↑)	C (Coupon ↓)	D (Yields ↓)
True duration	3.76	9.29	11.59	16.06
P^+	\$1,076.41	\$1,187.09	\$336.69	\$582.99
P_0	\$1,077.22	\$1,189.29	\$337.47	\$584.87
P^-	\$1,078.03	\$1,191.51	\$338.26	\$586.75
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True duration	3.76	9.29	11.59	16.06
P^+	\$1,000.00	\$1,000.00	\$272.74	\$434.41
P_0	\$1,077.22	\$1,189.29	\$337.47	\$584.87
P^-	\$1,162.22	\$1,452.47	\$434.41	\$826.20
PVBP	-0.81	-2.26	-0.81	-1.96
Eff. Duration	**3.76**	**9.51**	**11.98**	**16.75**

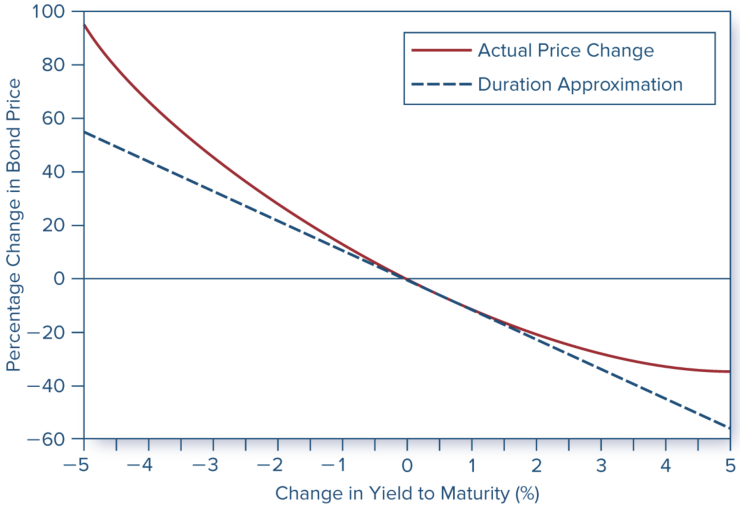
Caveat 1: What if Interest Rate Changes Are Large?

- There's a reason I've emphasized **small** changes in interest rates: it's only an approximation

- With long maturities, bond prices are highly **non-linear** in yields
  - Reduces the accuracy of our duration formula
  - (Think about calculus: a derivative is a *local* concept)

- **Previous exercise** highlights the importance of accounting for this non-linearity
  - Call this **convexity**

Visualizing Convexity and Duration Approximation Error



A Revised (But More Complicated) Interest Rate Risk Equation

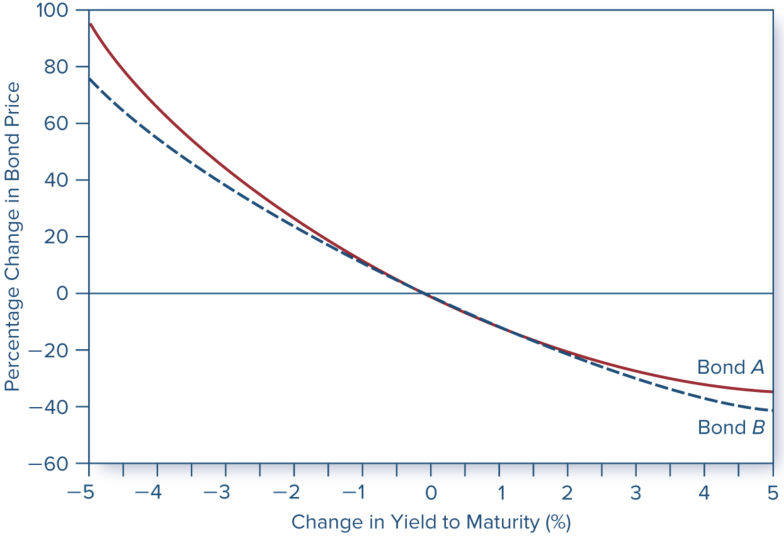
- We can adapt our previous equation to account for convexity:

$$\frac{\Delta P}{P} = -\boldsymbol{D} \times \frac{\boldsymbol{\Delta y}}{\boldsymbol{1+y}} + \frac{\boldsymbol{1}}{\boldsymbol{2}} \times \textbf{Convexity} \times (\boldsymbol{\Delta y})^2$$

- **Convexity** has a very complicated formula that I won't show here

- Why do we add in the term for convexity this way?
  - **Answer:** $\sim$ 2nd order Taylor approximation
  - When Δy is very small, the term basically goes away

Visualizing Convexity



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  1. *Increases* in yields lead to *smaller declines* in bond prices
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- Bonds with higher convexity therefore have **higher expected returns**

- **What do you think this implies for bond prices and yields?**

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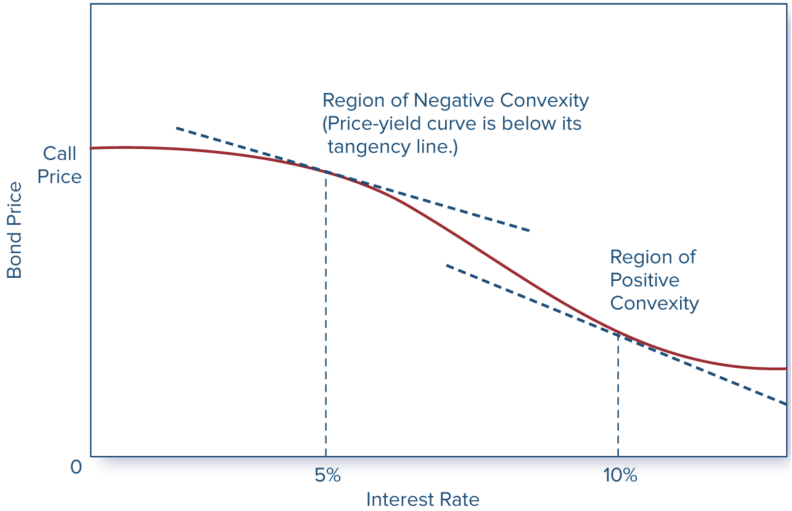
- Bonds with higher convexity therefore have **higher expected returns**

- **What do you think this implies for bond prices and yields?**
  - Answer: convex bonds tend to be priced higher with lower yields

What if we don't have an option-free bond?



How Do We Calculate Duration for Callable Bonds?

- We can't apply our formula from before due to the **embedded call option**
  - Cash flows are more complicated!

- **In practice:** directly calculate <u>effective duration</u> using prices and yields:

$$\textbf{Effective Duration} = -\frac{\Delta P/P}{\Delta r}$$

- **Note:** we can't use changes in yields as before! (Yield to maturity unknown)
  - Use *level* shifts in the **term structure** to approximate change in market rate
  - (Will discuss after the break)

Model and Duration Quality

- Even if we use effective duration, what assumptions are we making that might be misleading?

Model and Duration Quality

- Even if we use effective duration, what assumptions are we making that might be misleading?
- **A: We assume we know how to determine the price!**
- **Reality:** Callable bond prices often make modeling assumptions, so:
  - If model is bad, duration measure is bad
  - Can have large consequences!

Recap

- Today we discussed **duration**
  - Bond's interest rate sensitivity, but also effective cash flow maturity
  - Good estimate with small changes in yields, approximation otherwise
- Duration estimates useful for thinking about *hedging*
  - Will discuss in depth when we talk about interest rate swaps
- **After spring break:** Relax assumption on constant yields
  $\rightarrow$ **term structure of interest rates**